

The Effect of Medicaid Expansion on Health Insurance Coverage: Evidence and Heterogeneity Across Metro and Non-Metro Populations

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Abstract

This paper examines the effect of Medicaid expansion under the Affordable Care Act (ACA) on health insurance coverage and tests whether that effect differs across metro and non-metro populations. Prior research has established that Medicaid expansion increased coverage on average, but evidence also suggests that policy effects are not uniform across contexts. Using American Community Survey data obtained through IPUMS, this study applies a difference-in-differences framework comparing expansion and non-expansion states before and after the ACA Medicaid expansion. The main outcome is whether an individual has any health insurance coverage. To assess heterogeneity, the analysis incorporates metro status as an interaction with the treatment effect. This study contributes to the literature by examining whether the gains from Medicaid expansion differed by geographic context, with implications for understanding how large public insurance reforms affect populations facing different structural conditions.

1 Introduction

Access to health insurance is a central policy issue because it shapes whether individuals can obtain affordable and timely healthcare. In the United States, the Affordable Care Act (ACA) introduced Medicaid expansion as a major policy reform intended to increase insurance coverage, especially among low-income adults. Because states varied in whether they adopted the expansion, the policy created an important setting for studying the causal effects of public insurance reform.

A large body of research has shown that Medicaid expansion increased health insurance coverage overall. Existing studies also find broader improvements in access to care, financial protection, and selected health outcomes. Taken together, these results suggest that expanding Medicaid eligibility can generate meaningful gains in well-being.

At the same time, average treatment effects may conceal important differences across populations. Recent research indicates that the effects of Medicaid expansion are not uniform across contexts. In particular, studies of mortality find that the magnitude of the policy’s effects varies across states depending on demographic and structural characteristics. This broader evidence of heterogeneity suggests that public policy may operate differently across subpopulations that face different economic or healthcare environments.

One potentially important dimension of heterogeneity is geography. Individuals living in metro and non-metro areas may experience the benefits of Medicaid expansion differently because these settings differ in provider availability, healthcare infrastructure, transportation barriers, labor market structure, and baseline insurance conditions. Even if Medicaid expansion increases eligibility in both contexts, the extent to which that eligibility translates into actual insurance coverage may differ between metro and non-metro populations.

This paper asks: *Does Medicaid expansion increase health insurance coverage, and does this effect differ between metro and non-metro populations?* To answer this question, I use American Community Survey (ACS) data obtained through IPUMS and estimate a difference-in-differences model comparing expansion and non-expansion states before and after implementation of the ACA Medicaid expansion. I then extend the analysis by interacting the treatment effect with metro status to test for heterogeneous effects.

Consistent with the existing literature, Medicaid expansion is expected to increase the probability of having health insurance coverage. To capture this heterogeneity, the analysis includes a triple interaction between expansion status, the post period, and metro status. This interaction term measures whether the change in insurance coverage following Medicaid expansion differs between metro and non-metro populations, providing insight into whether policy effects are geographically uniform.

2 Literature Review

A large body of research has established that Medicaid expansion under the Affordable Care Act (ACA) increased health insurance coverage. Using quasi-experimental variation across states, Sommers et al. (2017) find that Medicaid expansion led to substantial reductions in uninsured rates, along with improvements in access to care and preventive service use. More recent evidence by Bin Abdul Baten et al. (2024) further confirms that Medicaid expansion

improved access to healthcare and influenced financial behavior among U.S. adults.

Beyond insurance coverage, Medicaid expansion has been shown to improve broader outcomes. Sommers et al. (2017) show that expansion is associated with increased access to a usual source of care, reductions in out-of-pocket spending, and improvements in self-reported health. These findings suggest that expanding coverage can have meaningful effects on both health and financial well-being.

At the same time, more recent research highlights that the effects of Medicaid expansion are not uniform across contexts. Lee et al. (2022) find that Medicaid expansion is associated with significant reductions in all-cause mortality, but that the magnitude of these effects varies substantially across states depending on demographic and structural characteristics. This variation indicates that policy impacts may depend on underlying contextual factors rather than operating uniformly across populations.

Taken together, the literature establishes three key findings. First, Medicaid expansion increased insurance coverage. Second, it improved broader outcomes related to access, financial protection, and health. Third, its effects are not uniform across populations and contexts.

While prior studies have documented overall increases in insurance coverage following Medicaid expansion, less attention has been paid to how these effects may differ across geographic contexts. Differences in healthcare infrastructure, provider availability, transportation barriers, and administrative access suggest that metro and non-metro populations may experience policy changes differently. This motivates the focus on geographic heterogeneity in the empirical analysis.

3 Policy Background

The ACA gave states the option to expand Medicaid eligibility to adults with incomes up to 138 percent of the federal poverty level. Although the law originally envisioned nationwide expansion, the 2012 Supreme Court decision in *NFIB v. Sebelius* made expansion optional for states. As a result, some states adopted Medicaid expansion beginning in 2014, while others did not.

This state-level variation is important for causal analysis. Because expansion occurred at a known point in time and not all states adopted it, the policy creates a quasi-experimental setting in which outcomes in expansion states can be compared to outcomes in non-expansion states before and after implementation. In this study, California is treated as an expansion state and Texas as a non-expansion state. California adopted Medicaid expansion, while Texas did not, making the pair a useful contrast for studying policy effects.

The choice to focus on insurance coverage reflects the core policy goal of Medicaid expansion. However, coverage gains may not be identical across places. Metro and non-metro areas differ in healthcare resources, economic conditions, and administrative access, all of which could affect whether a policy change translates into actual enrollment and coverage. For this reason, geographic heterogeneity is a meaningful extension of the standard analysis.

4 Data

The data for this analysis come from the American Community Survey (ACS) obtained through IPUMS. The ACS is a large, nationally representative survey that includes information on health insurance coverage, demographic characteristics, and geographic identifiers. Its large sample size makes it well suited for evaluating policy changes that vary across states and over time.

The sample is restricted to adults ages 19 to 64, since this is the population most directly affected by Medicaid expansion. Individuals age 65 and older are generally eligible for Medicare, so including them would weaken the connection between the policy and the outcome of interest.

The main outcome variable is an indicator for whether an individual has any health insurance coverage, constructed from `HCOVANY`. This binary measure captures whether the respondent is insured through any source and serves as the primary dependent variable in the analysis.

The treatment indicator is based on residence in an expansion state. In the current design, individuals living in California are classified as treated, while individuals living in Texas are classified as controls. The post-period indicator captures years after implementation of Medicaid expansion.

The analysis also uses several demographic covariates to improve comparability and precision. These include age, sex, and race. In addition, metro status is measured using `MET2013`, which is used to distinguish metro from non-metro populations. This variable is central to the heterogeneity analysis.

The sample period includes years before and after the 2014 ACA Medicaid expansion. Using multiple years on both sides of the policy change allows the analysis to compare changes over time rather than relying only on cross-sectional differences.

Table 1 reports summary statistics for the main variables used in the analysis, separately for California and Texas before and after Medicaid expansion.

Table 1 shows that the analysis sample consists of adults ages 19–64 in California and Texas observed before and after Medicaid expansion. Insurance coverage increases in both

Table 1: Summary Statistics

Variable	California		Texas	
	Pre (2013)	Post (2015)	Pre (2013)	Post (2015)
Insured	0.779 (0.415)	0.892 (0.310)	0.732 (0.443)	0.793 (0.405)
Age	41.569 (13.268)	41.729 (13.300)	41.685 (13.175)	41.910 (13.263)
Female	0.503 (0.500)	0.510 (0.500)	0.513 (0.500)	0.502 (0.500)
Metro	0.976 (0.153)	0.976 (0.152)	0.798 (0.401)	0.800 (0.400)
Observations	11934	12093	8035	8224

states over time, with a larger increase observed in California, the expansion state. Specifically, coverage rises from 77.9% to 89.2% in California, compared to an increase from 73.2% to 79.3% in Texas.

The demographic composition of the samples is broadly similar across states and periods, with average age around 42 and women comprising approximately half of each group. However, there are notable differences in geographic composition: nearly all California respondents reside in metro areas, compared to about 80% in Texas.

These descriptive patterns are suggestive of a potential effect of Medicaid expansion, but they should not be interpreted causally. California and Texas differ along multiple dimensions, including demographic and geographic characteristics, that may independently influence insurance coverage. A formal difference-in-differences framework is therefore required to estimate the causal effect of Medicaid expansion.

5 Empirical Strategy

5.1 Difference-in-Differences Framework

To estimate the effect of Medicaid expansion on insurance coverage, this paper uses a difference-in-differences (DiD) design. DiD compares the change in outcomes over time for a treated group to the change in outcomes over time for a control group. The intuition is that, by differencing over time and across groups, the design removes fixed differences between the groups and common time shocks that affect both groups.

In this study, the treated group consists of individuals residing in California, an expansion state, and the control group consists of individuals residing in Texas, a non-expansion state. The pre-period includes years before the implementation of Medicaid expansion, and the post-period includes years after implementation. The DiD estimate captures whether insurance coverage changed more in the expansion state than in the non-expansion state after the policy took effect.

The baseline specification is:

$$Y_{ist} = \alpha + \beta_1 \text{Expansion}_s + \beta_2 \text{Post}_t + \beta_3 (\text{Expansion}_s \times \text{Post}_t) + X'_{ist} \gamma + \varepsilon_{ist}$$

where Y_{ist} is an indicator for whether individual i in state s and year t has any health insurance coverage, Expansion_s indicates residence in the expansion state, Post_t indicates the post-expansion period, and X_{ist} is a vector of individual covariates including age, sex, and race. The coefficient of interest is β_3 , which represents the estimated effect of Medicaid expansion on insurance coverage.

5.2 Identification Assumptions

The key identifying assumption of the DiD approach is the *parallel trends assumption*. This assumption requires that, in the absence of Medicaid expansion, insurance coverage in the treatment and control groups would have evolved similarly over time. Under this assumption, any post-policy divergence between the groups can be attributed to the policy rather than to pre-existing differences in trends.

In this context, the parallel trends assumption is plausible but not guaranteed. California and Texas differ in important ways, including demographics, labor markets, and policy environments. These differences do not necessarily invalidate the design, but they do mean the assumption must be discussed carefully and, where possible, evaluated using pre-period trends.

A second consideration is that the treatment is exposure to the policy environment, not individual enrollment in Medicaid. Therefore, the DiD estimate should be interpreted as the effect of residing in a state that adopted Medicaid expansion, not as the effect of personally enrolling in Medicaid.

Because the analysis includes only one pre-treatment year, this assumption cannot be directly tested. Instead, the analysis relies on the plausibility of this assumption given similarities in observable characteristics and prior evidence in the literature. In this framework, the trend observed in Texas serves as an estimate of the counterfactual trend for California in the absence of expansion.

5.3 Heterogeneity by Metro Status

To examine whether the effects of Medicaid expansion differ across geographic contexts, I extend the baseline model to include metro status as a dimension of heterogeneity. Specifically,

I interact the DiD treatment effect with an indicator for metro status.

The heterogeneous-effects specification is:

$$Y_{ist} = \alpha + \beta_1 \text{Expansion}_s + \beta_2 \text{Post}_t + \beta_3 \text{Metro}_i \quad (1)$$

$$+ \beta_4 (\text{Expansion}_s \times \text{Post}_t) + \beta_5 (\text{Expansion}_s \times \text{Metro}_i) \quad (2)$$

$$+ \beta_6 (\text{Post}_t \times \text{Metro}_i) \quad (3)$$

$$+ \beta_7 (\text{Expansion}_s \times \text{Post}_t \times \text{Metro}_i) \quad (4)$$

$$+ X'_{ist} \gamma + \varepsilon_{ist} \quad (5)$$

In this model, β_7 captures whether the effect of Medicaid expansion differs between metro and non-metro populations. This interaction is important because it tests whether the policy had unequal effects across geographic settings rather than assuming a single average treatment effect.

6 Results

6.1 Overall Effects of Medicaid Expansion

Figure 1 presents changes in insurance coverage in California and Texas before and after Medicaid expansion. Insurance coverage increased in both states between 2013 and 2015, but the increase was larger in California, the expansion state. This pattern is consistent with the expectation that Medicaid expansion increased access to insurance.

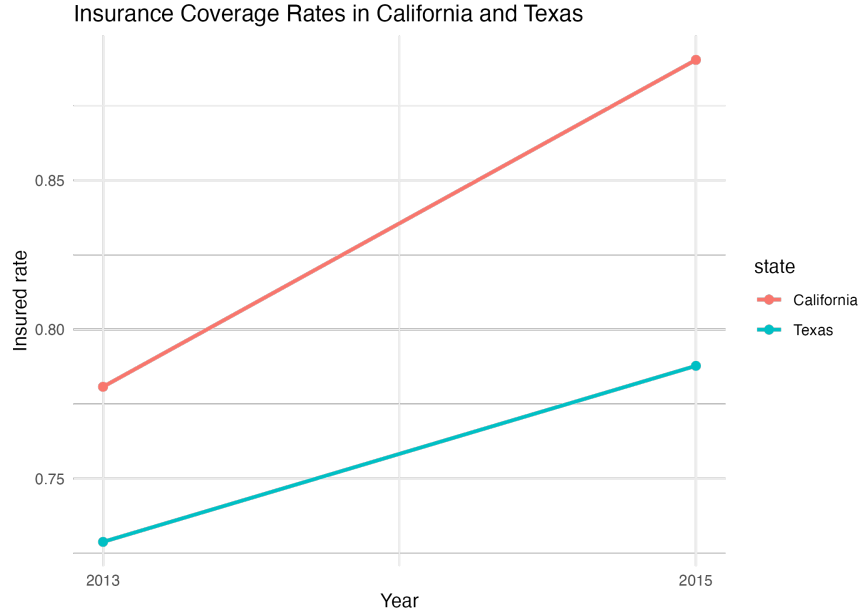


Figure 1: Insurance Coverage Rates in California and Texas

Table 2 reports the difference-in-differences regression results. The coefficient on the interaction between expansion status and the post period is positive, indicating that insurance coverage increased more in California than in Texas after Medicaid expansion. This provides empirical support for the first hypothesis and is consistent with findings in the existing literature.

6.2 Heterogeneity by Metro Status

Figure 2 shows changes in insurance coverage separately for metro and non-metro populations in both states. While coverage increased for both groups, the figure suggests slightly larger increases for metro populations.

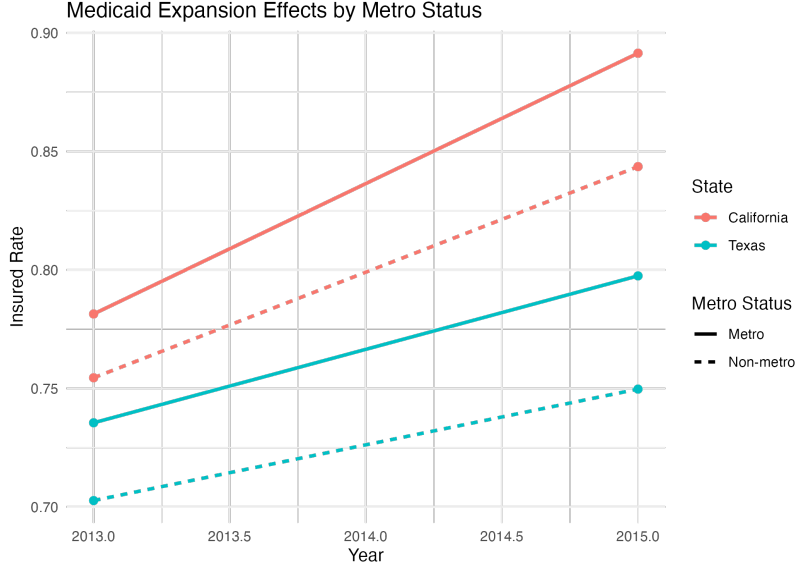


Figure 2: Medicaid Expansion Effects by Metro Status

However, regression results indicate that this difference is small and not statistically significant. The estimated triple interaction term ($\text{Expansion} \times \text{Post} \times \text{Metro}$) is 0.008 (SE = 0.009), implying that the effect in metro areas is only 0.8 percentage points larger, and this difference is not statistically distinguishable from zero.

Table 2: Difference-in-Differences Regression Results

	Main DiD	DiD x Metro
Expansion $\times$ Post	0.052*** (0.002)	0.042*** (0.008)
Metro		0.038*** (0.003)
Expansion $\times$ Metro		0.001 (0.007)
Post $\times$ Metro		0.014*** (0.004)
Expansion $\times$ Post $\times$ Metro		0.008 (0.009)
Observations	758,144	758,144
R^2	0.054	0.055

Robust standard errors in parentheses. *** p < 0.01

Although the graphical evidence suggests somewhat larger gains for metro populations, these differences are modest in magnitude and not statistically distinguishable from zero. This suggests that Medicaid expansion increased insurance coverage broadly across both metro

and non-metro populations rather than disproportionately benefiting one group.

6.3 Interpretation

Taken together, the results indicate that Medicaid expansion was effective in increasing insurance coverage, consistent with prior research. At the same time, the analysis does not provide strong evidence of heterogeneous effects across geographic contexts. While baseline differences in coverage persist between metro and non-metro populations, the magnitude of the policy’s impact appears to be relatively similar across these groups.

7 Conclusion

This paper finds that Medicaid expansion under the ACA significantly increased health insurance coverage, and that this effect was broadly similar across metro and non-metro populations. While the analysis does not find statistically significant differences in treatment effects across geographic contexts, this result is informative: expansion was effective in increasing coverage even in non-metro areas, where structural barriers such as provider shortages and limited administrative access might otherwise constrain enrollment.

However, similar treatment effects do not imply equal outcomes. Non-metro populations began with lower baseline coverage and remained at a disadvantage after expansion, indicating that uniform policy gains are insufficient to eliminate pre-existing geographic disparities. This suggests that expanding eligibility alone may not fully address gaps in access, and that additional policies such as targeted outreach, enrollment assistance, or administrative simplification may be necessary to further reduce disparities.

Together, these findings highlight the importance of distinguishing between equal policy effects and equal outcomes when evaluating large-scale health reforms.

8 Limitations

This analysis is subject to limitations related to identification, generalizability, and sample composition. First, the difference-in-differences approach relies on the parallel trends assumption, which cannot be directly tested due to the availability of only one pre-treatment year. As a result, the causal interpretation depends on the plausibility of this assumption.

Second, the analysis is based on a comparison between California and Texas, which may limit the generalizability of the findings. Differences between these states may influence the results, and the conclusions may not extend to other contexts.

Third, the distribution of metro and non-metro populations differs substantially across states, with California being predominantly metro. This imbalance limits the precision of the heterogeneity analysis and may contribute to the lack of statistically significant differences across groups.

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